Smart Greenhouse Monitoring System

IoT Project – First Semester 2025/2026

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1 Introduction

Project Motivation

- Need for continuous greenhouse monitoring
  - Temperature, humidity, and air quality affect plant growth
  - Manual monitoring is inefficient and unreliable
  - IoT enables real-time and accurate data collection

IoT technologies improve efficiency and decision-making in smart agriculture.

Problem Statement

- Manual monitoring lacks accuracy and reliability
- Sensor data is not easily accessible or centralized
- Inability to analyze long-term environmental trends

These limitations highlight the need for an automated and cloud-based IoT solution.

Proposed Solution

- Smart Greenhouse Monitoring System
  - ESP32-based IoT architecture
  - Real-time local display and cloud integration

The proposed system offers a reliable and scalable IoT solution for greenhouse monitoring.

System Capabilities

- Environmental Monitoring
  - Temperature and humidity measurement
  - Air quality (gas level) monitoring
- Data Presentation
  - Real-time LCD display
  - Cloud-based visualization and analysis

The system provides real-time insight into greenhouse environmental conditions.

2 Hardware and Software Implementation

System Architecture

1. Sensors collect environmental data
2. ESP32 reads and processes the data
3. Data displayed on a 20×4 LCD
4. Data sent to ThingSpeak via WiFi

System Simulation (Wokwi)

- Complete system simulated before hardware deployment
- Includes ESP32, DHT22, gas sensor, and 20×4 LCD
- Used to validate data flow and system behavior

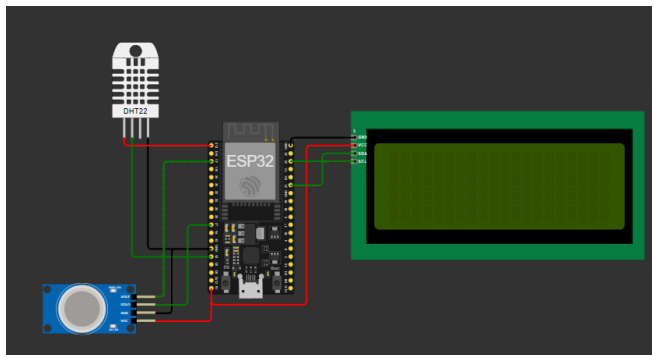

Figure 1: Wokwi simulation of the smart greenhouse monitoring system

Hardware Components

- Controller
  - ESP32 DevKit C V4
- Sensors
  - DHT22 (Temperature & Humidity)
  - Gas sensor
- Output
  - 20×4 I2C LCD display

These components form the core hardware of the monitoring system.

Hardware Connections

- DHT22 Sensor
  - Connected to GPIO 13
- LCD Display (I2C)
  - SDA ⟶ GPIO 21
  - SCL ⟶ GPIO 22
- Gas Sensor
  - Provides analog output
  - Provides digital output

Proper wiring ensures reliable sensor readings and data transmission.

Software Implementation

- Development Environment
  - Arduino framework (C++)
- Data Processing
  - Sensor readings every 2 seconds
  - Heat index calculated from temperature and humidity
  - Gas levels classified using predefined thresholds
- Reliability
  - Error handling for sensor failures

The software ensures reliable data processing and real-time system operation.

ThingSpeak Cloud Platform

- Platform Role
  - Cloud storage for sensor data
  - Data visualization and analysis
- Data Field Mapping
  - Field 1: Temperature
  - Field 2: Humidity
  - Field 3: Gas sensor value
  - Field 4: Heat index
  - Field 5: Gas classification

ThingSpeak Cloud Platform (ii)

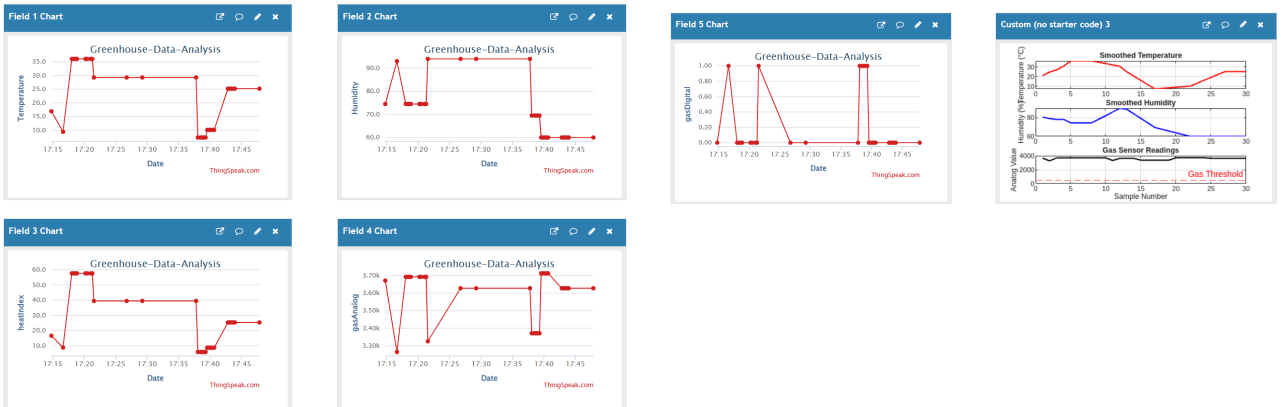

Figure 2: ThingSpeak dashboard and MATLAB-based visualization for greenhouse data analysis

3 Results and Discussion

Results

- System Performance
  - Accurate and reliable sensor readings
  - Continuous real-time monitoring
- Data Insights
  - Improved accessibility to environmental data
  - Support for long-term trend analysis

The results confirm the effectiveness of the proposed IoT-based monitoring system.

4 Conclusion and Future Work

Conclusion

- Successful implementation of a smart greenhouse monitoring system
- Accurate monitoring of key environmental parameters
- Integration of local and cloud-based visualization
- Suitable for smart agriculture applications

The proposed system demonstrates the effectiveness of IoT technologies in smart agriculture.

Future Work

- System Expansion
  - Integration of additional sensors (e.g., soil moisture)
- Automation
  - Automated irrigation and ventilation control
- User Interface
  - Mobile application integration
- Advanced Analytics
  - Predictive analytics using machine learning

These enhancements can further improve efficiency and intelligence in smart greenhouse systems.

5 Deliverables

Project Deliverables

- Wokwi Simulation: https://wokwi.com/projects/450157931815022593
- ThingSpeak Channel (Monitoring): https://thingspeak.mathworks.com/channels/3201893
- ThingSpeak Channel (Webhook / Action): https://thingspeak.mathworks.com/channels/3209603

6 End

Thank You

Questions?